

Mitigating Baseline Shift in High Event-Rate Systems via Pole-Zero Cancellation and Wiener Filtering

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Abstract—Energy reconstruction in high-energy calorimetry systems is strongly affected by signal conditioning, especially under high event-rate conditions. Shaping stages introduce long-duration components in the pulse response, leading to baseline shifts and degraded estimation performance. This work investigates the use of Pole-Zero Cancellation (PZC) as a pre-processing stage combined with Wiener filtering for amplitude estimation. The performance of the Wiener filter alone is compared with the PZC + Wiener approach. The PZC stage compensates for the exponential component introduced by the readout chain, stabilizing the baseline prior to reconstruction. Results show that, while both approaches perform similarly under low-occupancy conditions, the Wiener filter alone degrades significantly in high-occupancy scenarios due to systematic bias. The proposed PZC + Wiener approach reduces reconstruction error and improves robustness in environments with significant signal overlap.

Index Terms—energy reconstruction, Wiener filter, pole-zero cancellation, digital signal processing, baseline correction, high event-rate systems

I. INTRODUCTION

The study of high-energy particle collisions enables the investigation of fundamental interactions of matter by analyzing signals generated in specialized detectors (1). To explore the discovery potential of high-luminosity collider scenarios, the development and validation of new acquisition algorithms rely heavily on advanced computer simulations. In this context, platforms such as *Lorenzetti Showers* (2) stand out as essential tools for signal modeling and algorithm evaluation.

Lorenzetti is a general-purpose, open-source framework designed to support studies on trigger algorithms and signal reconstruction in segmented calorimeters. It provides detailed information on the readout values of computational cells, allowing the configuration of different pulse shapes and the modeling of realistic experimental conditions involving signal

and noise superposition. This flexibility enables the reproduction of a wide range of operating regimes, including scenarios with different levels of noise and event density.

Following the digitization of the waveform generated in the readout channels, the signals are represented by a finite set of time samples. These samples contain not only the contribution of the event of interest but also electronic noise and the overlapping of adjacent events, known as pile-up. In this context, energy reconstruction is formulated as the estimation of pulse amplitudes from noisy and distorted observations, being directly influenced by both the statistical properties of the signal and the characteristics of the readout chain (3).

In high event-rate environments, the occurrence of pile-up becomes severe, resulting in the superposition of multiple signals within the same acquisition window. This condition alters the observed pulse shape, introduces temporal correlations, and significantly degrades the accuracy of direct amplitude estimation. Moreover, the signal conditioning stage typically introduces long-duration exponential components, which persist over time and accumulate as new events are recorded. As a consequence, the signal baseline becomes unstable, leading to systematic deviations in the reconstructed amplitude.

In this scenario, techniques based on statistical filtering, such as the Wiener filter, provide a suitable framework for amplitude estimation, as they explicitly account for the correlation structure of the signal and noise (4). By minimizing the mean squared error, the Wiener filter yields an optimal linear estimator under stationary conditions. However, its performance can degrade in the presence of baseline shifts and low-frequency distortions, which violate the assumptions of stationarity and zero-mean behavior typically required by the model.

To address this limitation, (5) proposed a Pole-Zero Can-

cellation (PZC) technique capable of operating in a streaming context under high-occupancy conditions. This method compensates for the exponential component responsible for the long-duration tail introduced by the shaping stage, effectively reducing the memory of the system and restoring baseline stability.

This work proposes the combined use of a baseline correction stage based on the PZC technique, followed by Wiener filtering for energy estimation. The approach is validated using synthetic data generated by the *Lorenzetti* framework, and its performance is evaluated under different occupancy conditions. The results demonstrate that the proposed architecture improves robustness and reduces reconstruction error in scenarios with significant signal overlap.

The remainder of this work is organized as follows: Section II formalizes the energy reconstruction problem and describes amplitude estimation using the Wiener filter. Section III details the characteristics of the simulated signal shaping and explains the origin of baseline shift. Section IV presents the PZC technique as a pre-processing strategy. Section V presents simulation results and performance evaluation. Finally, Section VI summarizes the conclusions.

II. ENERGY RECONSTRUCTION

In high-energy calorimetry systems, the energy deposited by particles is inferred from electrical signals generated in the readout chain. After the interaction in the detector, the signal is processed by the front-end electronics, which include shaping stages responsible for defining an approximately deterministic pulse shape. As a result, the pulse amplitude becomes proportional to the deposited energy, allowing the estimation of this quantity from digitized samples.

However, in high event-rate environments, such as those expected in next-generation collider scenarios, the temporal overlap of signals from different interactions occurs, a phenomenon known as pile-up. This effect distorts the observed pulse shape and compromises direct amplitude estimation, making it necessary to employ digital signal processing techniques for energy reconstruction (6).

In this context, the acquisition system can be modeled as a linear time-invariant system, in which the observed signal corresponds to the convolution between the sequence of amplitudes associated with events and the impulse response of the readout chain, in addition to noise:

$$y[n] = (a * s)[n] + \eta[n] = \sum_m a[m] \cdot s[n - m] + \eta[n], \quad (1)$$

where $s[n]$ represents the system pulse shape, $a[n]$ is the sequence of amplitudes associated with interactions over time, $*$ denotes the discrete convolution operation, and $\eta[n]$ corresponds to electronic noise. This model is widely used in the literature to describe the pile-up effect as a superposition of time-shifted responses (7).

Energy reconstruction therefore consists of estimating the sequence $a[n]$ from the observed signal $y[n]$. In online implementations, this task is typically performed using linear

FIR filters, in which the estimate is obtained from a weighted combination of consecutive samples:

$$\hat{a}[n] = \sum_{k=0}^{N-1} w[k] \cdot y[n - k], \quad (2)$$

where $w[k]$ are the filter coefficients and N defines the number of samples considered in the estimation.

The determination of these coefficients can be formulated as an optimization problem. Among the possible approaches, the Wiener filter stands out, as it seeks to minimize the mean squared error between the estimated amplitude and the true value (8; 9). In this case, the optimal solution can be expressed as

$$\mathbf{w} = \mathbf{R}^{-1} \mathbf{p}, \quad (3)$$

where $\mathbf{R}$ is the autocorrelation matrix of the observed signal and $\mathbf{p}$ is the cross-correlation vector with the desired sequence (9).

In practice, these parameters are estimated from simulated or experimental data using a least-squares formulation. For this purpose, an observation matrix $\mathbf{Y}$ is constructed from the signal $y[n]$, organized in temporal windows, where each row contains N consecutive samples corresponding to the input window of the FIR filter. Thus, the number of filter coefficients is directly related to the number of columns of $\mathbf{Y}$. The weight vector can then be obtained as

$$\hat{\mathbf{w}} = (\mathbf{Y}^T \mathbf{Y})^{-1} \mathbf{Y}^T \mathbf{a}, \quad (4)$$

where $\mathbf{a}$ is the vector of desired amplitudes (9).

Despite the robustness of this approach, its performance can degrade significantly in the presence of systematic distortions. In particular, low-frequency components and baseline shifts introduced by the shaping stage violate the assumption that the observed signal is centered around a stable reference level. As a result, the estimation error becomes dominated by bias rather than by stochastic noise, leading to inaccurate amplitude reconstruction even when the filter is optimally designed.

These limitations become especially critical under high event-rate conditions, where the accumulation of residual signal components further amplifies baseline instability. In such scenarios, additional signal conditioning strategies become necessary to restore the conditions required for reliable amplitude estimation.

III. SHAPING CIRCUIT

As discussed in the previous section, energy reconstruction depends directly on the quality of the signal provided to the digital processing stage. In high-energy calorimetry simulations, such as those supported by the *Lorenzetti Showers* framework, the signal is not treated as an ideal impulse. Instead, it results from mathematical models that represent the front-end electronics readout chain (2), including amplification, shaping, and subsequent digitization.

As illustrated in Figure 1, synthetic signals corresponding to energy deposition events pass through computational routines that reproduce the analog signal conditioning stages, followed

by discretization that emulates the resolution and sampling frequency of analog-to-digital converters (ADCs).

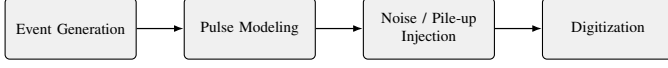


Fig. 1: Diagram illustrating the signal-generation and readout routines implemented in the *Lorenzetti Showers* framework.

In this context, accurate modeling of the front-end signal conditioning is essential for defining the temporal characteristics of the waveform. Beyond scaling the pulse amplitude to the system dynamic range, the shaping stage directly determines its evolution in time, influencing the sampled waveform and, consequently, the performance of the estimation methods applied in the digital domain.

The temporal response of the signal after the shaping stage is governed by the characteristics of the equivalent analog model. In particular, the shaper can be represented as a network of passive elements that introduces poles and zeros into the system transfer function (10). A commonly adopted representation consists of a cascade of differentiator and integrator stages, known as a CR- n RC structure, which provides an effective compromise between noise suppression and pulse shaping (11).

Assuming identical time constants for all stages, the transfer function of a CR-4RC shaping circuit can be expressed as

$$H(s) = \frac{CRs}{(CR)^5s^5 + 9(CR)^4s^4 + 28(CR)^3s^3 + 35(CR)^2s^2 + 15CRs + 1} \quad (5)$$

which describes the main temporal characteristics of the shaping stage and captures the pulse broadening introduced by the signal conditioning process (5; 10).

The corresponding impulse response exhibits a characteristic waveform, consisting of a fast rise followed by a slower decay. This behavior is determined by the system time constants, which define the pole distribution of the transfer function. As a consequence, the shaped pulse does not return immediately to the baseline after its peak, but instead presents a long-duration residual component.

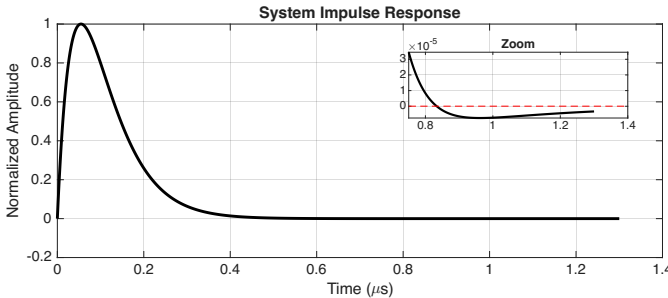


Fig. 2: Impulse response of the shaping stage, highlighting the presence of a long-duration tail component.

Although the amplitude of this tail is significantly smaller than that of the main peak, it persists over a considerably

longer time interval, as shown in Figure 2. This long-tail effect becomes particularly relevant in high event-rate conditions, where residual contributions from previous pulses may still be present when subsequent events are recorded.

Under low-occupancy conditions, this effect has limited impact on reconstruction, since the signals remain sufficiently separated in time. However, as the event rate increases, the temporal overlap between pulses becomes more frequent. In this regime, the residual decay components accumulate over time, leading to a gradual displacement of the signal baseline (5).

This behavior can be observed in the time domain, where the accumulation of residual components leads to a progressive deviation from the reference level, as illustrated in Figure 3. In such conditions, the signal no longer oscillates around a stable baseline, but instead exhibits variations that depend on the event rate and on the temporal history of the system.

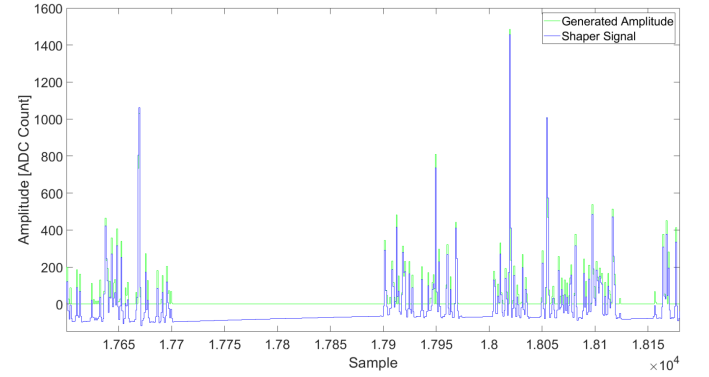


Fig. 3: Example of signal behavior under high-occupancy conditions, illustrating the baseline shift caused by the accumulation of long-duration pulse components.

This effect is further aggravated by the presence of noise and low-frequency components introduced by the readout chain, which are not completely suppressed by the shaping process.

From the perspective of energy reconstruction, this behavior introduces a systematic error in amplitude estimation. Linear filtering methods, such as the Wiener filter, typically assume that the input signal is centered around a stable reference level. When this assumption is violated, the estimation process becomes biased, leading to significant deviations from the true amplitude.

Therefore, within the simulated environment provided by *Lorenzetti*, the shaping stage plays a central role in defining the distortions that must be addressed prior to amplitude estimation. This motivates the use of additional pre-processing techniques capable of compensating for the long-duration component and restoring baseline stability, as discussed in the next section.

IV. PZC IN AN ONLINE ENVIRONMENT

As discussed in the previous section, the shaping circuit response introduces a long-duration decay component into the signal, which, under high event-rate conditions, leads

to baseline shifts and degradation of energy reconstruction. These effects are associated with the presence of poles in the system transfer function, which give rise to the exponential tail observed in the pulse.

An effective approach to mitigate this issue is the application of the Pole-Zero Cancellation (PZC) technique (5). This method operates by compensating the system poles through the introduction of appropriately placed zeros, attenuating the exponential component and promoting a faster return to the baseline.

In this work, PZC is employed as a pre-processing stage applied to the digitized signal, acting directly on the shaping circuit response (Figure 4). This approach reduces the influence of the long-duration component and restores the signal to a behavior closer to a short-duration pulse.

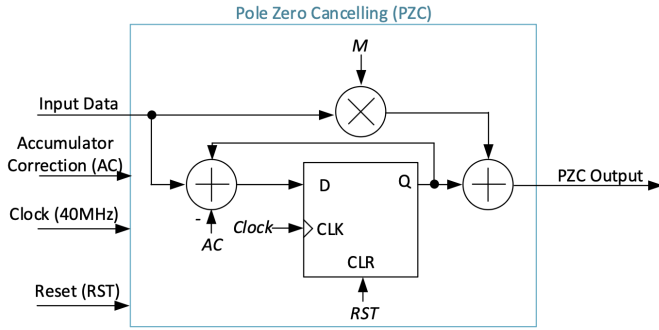


Fig. 4: Pole-Zero Cancellation (PZC)

The digital implementation of PZC can be described by a first-order recursive filter. For an input signal $v[n]$, the output $r[n]$ is given by:

$$\begin{aligned} p[n] &= v[n] + p[n-1], \\ r[n] &= M \cdot v[n] + p[n], \end{aligned} \quad (6)$$

where $M = (e^{T_s/\tau} - 1)^{-1}$, with τ representing the time constant of the exponential component and T_s the sampling period (5).

This formulation reveals that the PZC operates as a discrete-time compensation of the exponential decay introduced by the shaping circuit. While the shaping stage produces a response dominated by a pole associated with the long-duration component, the PZC introduces a corresponding zero designed to cancel this effect. As a result, the overall system response exhibits a significantly reduced memory, allowing the signal to return more rapidly to its baseline.

This behavior can be interpreted as a temporal whitening of the signal baseline, in which the low-frequency components associated with the exponential decay are attenuated. Consequently, the signal becomes less dependent on its past values, reducing the correlation between consecutive samples introduced by the shaping process. This property is particularly relevant in high-occupancy conditions, where the accumulation of residual components from previous events leads to persistent distortions in the observed waveform.

The choice of the parameter M is critical for the performance of the method. Inadequate values result in incomplete tail cancellation or overcompensation, introducing distortions into the signal. In practice, this parameter can be obtained through system characterization or adjusted empirically according to the operating conditions.

From a computational perspective, PZC presents low complexity, involving only operations between consecutive samples, which enables its implementation in real-time digital systems. By limiting the accumulation of residual energy between consecutive events, the method mitigates signal overlap effects under high event-rate conditions.

V. RESULTS

To evaluate the performance of the proposed approach, a comparison was conducted between energy reconstruction using only the Wiener filter and the combined PZC + Wiener approach. The analyses were carried out considering two distinct scenarios: low occupancy and high occupancy, representing, respectively, near-ideal conditions and regimes with severe signal overlap.

A. Low Occupancy

Initially, a low-occupancy scenario was considered, in which the occurrence of pile-up is reduced and the signals are sufficiently separated in time, resulting in a low probability of overlap. Under these conditions, the effects associated with the long-duration tail and baseline shift are minimal, allowing evaluation under near-ideal conditions.

Figure 5 presents the time-domain reconstruction for the different methods using 15 filter coefficients.

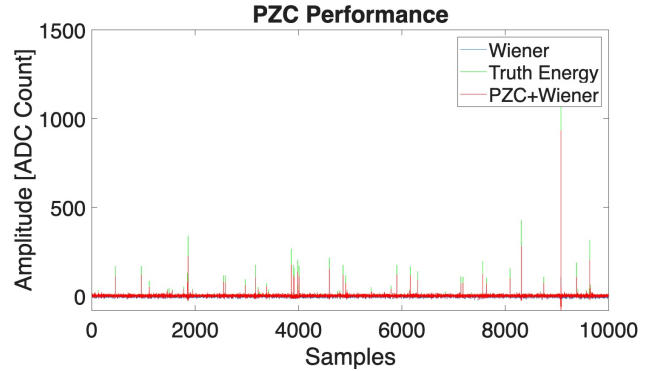


Fig. 5: Signal reconstruction in the time domain under low-occupancy conditions, comparing the reference signal, the Wiener filter, and the PZC + Wiener approach.

In this regime, the baseline remains approximately stable and the signals are well-defined, resulting in very similar amplitude estimates across the evaluated approaches.

The histograms presented in Figure 6 confirm this behavior, showing similar distributions for both methods. The errors exhibit an approximately symmetric distribution centered around zero, indicating the absence of significant bias in the amplitude estimation. Furthermore, the nearly Gaussian shape of the

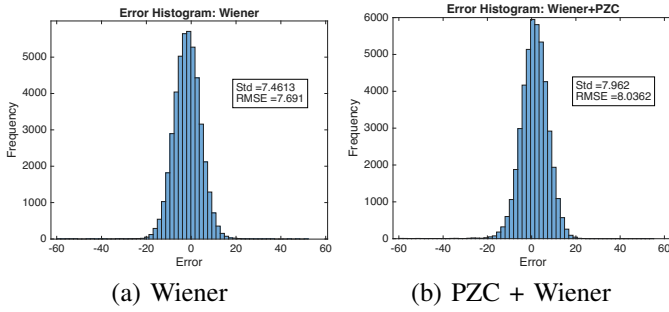


Fig. 6: Reconstruction error histograms under low-occupancy conditions.

distributions suggests that the error is predominantly governed by the statistical noise of the system.

The differences between the methods are small and consistent with statistical fluctuations, as reflected by the close values of standard deviation and RMSE. This indicates that, under low-occupancy conditions, the assumptions underlying the Wiener filter model are approximately satisfied. In this regime, the absence of significant baseline shifts and the low level of signal overlap result in amplitude estimation being dominated by noise, leading to similar performance across the evaluated approaches.

To complement this analysis, Table I presents the error metrics for different numbers of filter coefficients, allowing a comparison of the methods as a function of the number of weights used.

TABLE I: Reconstruction performance under low-occupancy conditions

Weights	Wiener		PZC + Wiener	
	Std	RMSE	Std	RMSE
03	7.37 ± 0.69	7.58 ± 1.18	8.59 ± 1.18	8.83 ± 1.13
05	7.37 ± 0.69	7.57 ± 0.69	8.31 ± 1.08	8.44 ± 1.05
07	7.37 ± 0.69	7.57 ± 0.69	8.14 ± 1.00	8.23 ± 0.98
09	7.37 ± 0.69	7.57 ± 0.69	8.01 ± 0.94	8.06 ± 0.92
11	7.37 ± 0.69	7.57 ± 0.69	7.92 ± 0.90	7.96 ± 0.89
13	7.37 ± 0.69	7.57 ± 0.69	7.85 ± 0.87	7.88 ± 0.86
15	7.37 ± 0.69	7.57 ± 0.69	7.80 ± 0.86	7.82 ± 0.84

The results in Table I reinforce this analysis, showing that the performance metrics are very similar for both approaches, as expected in a regime where pile-up effects and baseline shifts are negligible.

B. High Occupancy

A high-occupancy scenario was then considered, representing conditions with significant pile-up, where multiple events contribute simultaneously to the observed signal. In this regime, pulse overlap and the presence of the long tail lead to the accumulation of residual components over time, resulting in baseline shifts.

Figure 7 presents the time-domain reconstruction considering a configuration with 15 filter coefficients:

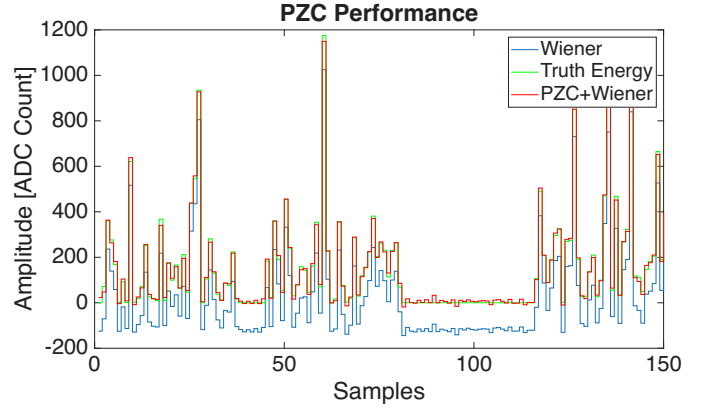


Fig. 7: Signal reconstruction in the time domain under high-occupancy conditions, comparing the reference signal, the Wiener filter, and the PZC + Wiener approach.

It can be observed that, in the absence of pre-processing, the signal reconstructed using the Wiener filter exhibits distortions associated with baseline shifts, leading to amplitude estimates affected by residual components. In contrast, the application of PZC stabilizes the baseline, allowing for a better agreement with the reference signal.

The histograms in Figure 8 highlight significant differences between the approaches:

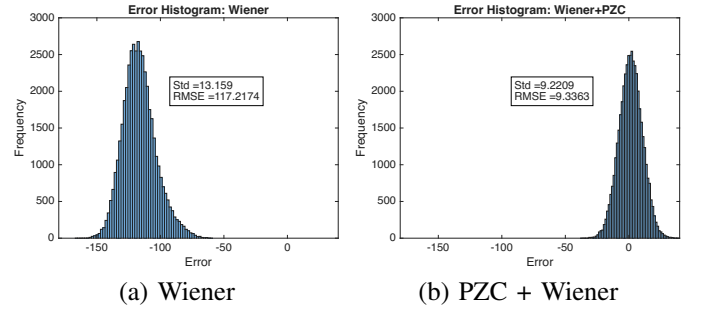


Fig. 8: Reconstruction error histograms under high-occupancy conditions.

The Wiener filter alone exhibits a clearly asymmetric distribution that is shifted away from zero, indicating the presence of a strong systematic bias. This behavior is primarily caused by the baseline displacement introduced by the long-duration tail, which leads to persistent overestimation or underestimation of the signal amplitude depending on the local accumulation of residual components.

In addition to the shift in the mean value, the asymmetry of the distribution reflects the non-stationary nature of the signal under high-occupancy conditions. Due to the superposition of multiple events and the accumulation of residual components, the estimation error is influenced by the temporal history of the signal, deviating from a symmetric, noise-limited regime.

On the other hand, the PZC + Wiener approach produces a distribution that is approximately symmetric and centered around zero, indicating the effective removal of the systematic

TABLE II: Reconstruction performance under high-occupancy conditions for different numbers of filter coefficients

Weights	Wiener		PZC + Wiener	
	Std	RMSE	Std	RMSE
03	17.82 ± 0.54	120.30 ± 2.56	14.70 ± 0.32	14.70 ± 0.31
05	14.11 ± 0.40	119.85 ± 2.55	10.56 ± 0.13	10.97 ± 0.13
07	13.75 ± 0.39	119.79 ± 2.55	9.76 ± 0.09	10.03 ± 0.11
09	13.69 ± 0.39	119.78 ± 2.54	9.53 ± 0.09	9.74 ± 0.10
11	13.66 ± 0.40	119.78 ± 2.55	9.40 ± 0.08	9.57 ± 0.09
13	13.56 ± 0.39	119.76 ± 2.54	9.36 ± 0.09	9.52 ± 0.09
15	13.56 ± 0.39	119.76 ± 2.54	9.34 ± 0.09	9.52 ± 0.09

bias. This behavior demonstrates that the baseline stabilization provided by PZC restores the conditions under which the Wiener filter assumptions are approximately satisfied. Furthermore, a significant reduction in error dispersion is observed, confirming that the estimation process returns to a regime predominantly limited by stochastic noise.

The results in Table II confirm this trend. The Wiener filter alone presents high error values that remain nearly constant for different numbers of coefficients, highlighting its limitation under high-occupancy conditions. This behavior is associated with the presence of significant bias in the estimation due to baseline shift.

In contrast, the PZC + Wiener approach leads to a substantial reduction in error metrics, with significantly lower RMSE values. This result demonstrates that baseline correction is essential to restore appropriate estimation conditions, making the method more robust in high event-rate environments.

VI. CONCLUSION

This work presented a comparative analysis between the Wiener filter and the combined PZC + Wiener approach for energy reconstruction, considering conditions representative of high event-rate environments. Under low-occupancy scenarios, both approaches exhibited similar performance, as the assumptions of the estimation model are approximately satisfied and the error is predominantly governed by statistical noise.

In high-occupancy regimes, however, a significant degradation in the performance of the Wiener filter was observed when used in isolation. This degradation is associated with the presence of long-duration components introduced by the shaping circuit, which lead to baseline shifts and result in systematic errors in amplitude estimation.

The application of PZC as a pre-processing stage proved effective in compensating for these effects, promoting bias removal and baseline stabilization. As a consequence, the PZC + Wiener approach achieved a significant reduction in reconstruction errors and improved robustness under high event-rate conditions.

The obtained results indicate that the PZC + Wiener combination constitutes a simple and efficient solution for energy reconstruction in environments with high pile-up, making it particularly suitable for applications in scenarios with significant signal overlap.

As future work, an FPGA implementation of the PZC + Wiener approach could be developed to assess its performance in a real-world energy reconstruction task.

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